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Experimental Implementation of Shor's Quantum Algorithm to Break RSA

Aminah Albuainain
College of Computer Science and Information Technology
Imam Abdulrahman bin Faisal University
Khobar, Kingdom of Saudi Arabia
2190003367@iau.edu.sa

Jana Alansari
College of Computer Science and Information Technology
Imam Abdulrahman bin Faisal University
Khobar, Kingdom of Saudi Arabia
2190004229@iau.edu.sa

Samiyah Alrashidi
College of Computer Science and Information Technology
Imam Abdulrahman bin Faisal University
Khobar, Kingdom of Saudi Arabia
2190004831@iau.edu.sa

Wasmiyah Alqahtani
College of Computer Science and Information Technology
Imam Abdulrahman bin Faisal University
Khobar, Kingdom of Saudi Arabia
2190001532@iau.edu.sa

Jana Alshaya
College of Computer Science and Information Technology
Imam Abdulrahman bin Faisal University
Khobar, Kingdom of Saudi Arabia
2190004722@iau.edu.sa

Naya Nagy
College of Computer Science and Information Technology
Imam Abdulrahman bin Faisal University
Khobar, Kingdom of Saudi Arabia
nmnagy@iau.edu.sa

***Abstract*— Nowadays, quantum computing is an important topic since it allows computations to be completed in a small amount of the time. The goal of using quantum computers (QC) is to solve an increasing number of problems that were previously intractable to address with traditional computing. The classical approaches work with bits, which are made up of 0s and 1s, whereas QC uses qubits. Classical computing is limited by storage capacity and speed of calculations, even when parallel computation is conducted on it. Compared to traditional methods, quantum parallelism allows the storage to be reduced exponentially with the potential of a shorter amount of time. Besides, Peter Shor's algorithm can solve the factorization problem used in RSA in polynomial time, whereas in a classical computer, factoring any big integer is intractable. The purpose of this research is to explore and implement quantum computing schemes and how Shor's algorithm can break RSA algorithms.**

Keywords*— Quantum Computing, Quantum Phase Estimation, Quantum Fourier Transformation, Hadamard Gate, Factorization

I. Introduction

Quantum computers (QC) were introduced in 1994 to operate computations using computers. It is a neoteric technology that uses quantum physics to resolve problems that standard computers cannot obtain. The main purpose for developing quantum computers was to deal with enormous amounts of data by employing the principles of entanglement, interference, and superposition of qubits and running them simultaneously. These technologies are not like the standard computers that have been around for 50 years, quantum algorithms take a suitable perspective to these kinds of problems by creating multidimensional spaces in which patterns linking individual data points emerge. Traditional computers are not able to realize these patterns because they are unable to create new computational regions. As a real example, traditional computer could not find folding patterns in proteins, while early quantum algorithm can find it in more effective ways, without the time-consuming checks that classical computers need [1]. In QC, many different algorithms have been created: the first is the Deutsch Jozsa Algorithm, it was to compete with classical computers; the second is Grover's Algorithm, is used in information extraction and works with unstructured data; the third is Simon's Algorithm, it was optimizing the computation process; and finally, Shor's Algorithm. It was introduced by Peter Shor to solve the factorization problem, and its purpose is to represent the algorithm's factors in polynomial time. On the other hand, the advent of quantum computing poses a threat to contemporary cryptography, as most security specialists are aware. Shor's quantum method, especially, provides a significant theoretical speedup to attackers attempting to brute-force several public-key cryptosystems like RSA and ECDSA [2]. The discovery of various quantum algorithms has posed a severe challenge to modern encryption, with Shor's algorithm currently being the most serious threat to cryptanalysis. RSA is an effective algorithm that is developed based on using two keys, a public key, and a private key, where public key is used for encryption and the private key for decryption. It is built on the idea of prime factorization, which is simple to break when the text size is small, and it will become harder to break as the text size grows. Shor's approach for attacking RSA is based on solving the Integer Factorization Problem (IFP), although breaking RSA does not have to be solved by solving the IFP. A quantum technique is being developed to attack the RSA cryptosystem from the non-factorization perspective. That is,

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RSA is inoperable unless the IFP is solved. Unlike prior attempts by cryptanalysts to recover the private key, a ciphertext-only attack technique for RSA is described, starting with retrieving the plaintext M [3]. The present work was designed to differentiate between the complexity of brute force attack against RSA with classic computer and quantum computers which have significant reduction from exponential to polynomial. Finally, the intended goal is to suggest a countermeasure and possibility for such quantum attack.

II. PROBLEM STATEMENT

This study explores the possibility of breaking RSA using Shor's algorithm with a proof-of-concept implementation on a quantum computer, the importance of this endeavor stems from the fact that RSA is broadly used in traditional banking systems and for digital signatures. However, due to continuous quantum computing improvements, current security systems may be endangered. Classical computers run factorization algorithms in exponential running time which makes it impossible to crack a large number. By contrast, quantum computers using Shor's algorithm perform factoring of big integers in polynomial time [4].

III. LITERATURE OF REVIEW

Bhatia & Ramkumar [5] review Quantum Computing Algorithms and how Shor's algorithm can break the RSA scheme. Their major challenge in developing their methodology was determining the value period, which was handled utilizing quantum computing and Shor's algorithm. At the end of their algorithm, they use the Euclidean Algorithm to calculate the period of the equation. After implementing the proposed algorithm with Qiskit, they conclude that there are several models with the number of bits mapped against runtime on the reproduced graph. Fedele & Asaithambi [6] created a computer code that reproduces Shor's algorithm for obtaining prime numbers. Their method begins with a classical step of selecting a random number and finding its relatively prime using Euclid's algorithm, followed by a quantum computation that uses a period-finding subroutine to identify the period r of the method, f(x)=ax mod (N), and ends with a classical step of extracting the period by mathematical manipulation. They conclude their work by comparing it to a standard classical factoring algorithm, and their implementation of the Shor's algorithm results in the number 123, since the only available qubit number is 32. Sharma et al. [7] main goal is examining prime integer factorization to uncover vulnerabilities in the usage of RSA encryption as a manner of security. To achieve this, authors first built a simple Boolean logic circuit to find multiplication of two numbers and show circuit's flow. Then, reverse it by employing the constraint satisfaction problem. Then they program one logic gate at a time into the quantum computer and map the qubits and couplers by applying transformations to all the gates in the circuit then put it to the test on a quantum computer. Due to this examination the result reveals that RSA encryption can be cracked. Nene & Upadhyay [8] proposed a systematic way to factor integers on a traditional system utilizing Shor's quantum algorithm. Platforms for simulation and the algorithm have been mentioned. The results reveal that the factoring issue can be solved on a traditional computer, and the results are validated using Shor's quantum factoring algorithm and basic simulation. However, there are certain limitations because registers in typical computers cannot be in superposition. The second and most significant issue is that as N goes greater, order finding, r, in the simulation becomes harder. Thombre & Jajodia [4] provides an overview of RSA, Rabin, and Shor algorithms. furthermore, shows how to attack RSA and Rabin algorithms using Shor's algorithm by breaking the integer factorization. However, due to the constraints of 32-qubits in current IBM quantum computers, the experimental examination of Quantum Shor's algorithm on existing public-key cryptosystems utilizing IBM Quantum Experience is undertaken as a proof of concept for factorizing integers of short length (7-bits). In a sentence, this paper shows how Shor's algorithm puts secrecy and authentication services at risk. Ugwuishiwu et al. [9] looking for the mechanisms of quantum cryptography and comparing quantum and classical encryption techniques. The authors were able to concisely introduce quantum cryptography, clarify how encryption is done by utilizing quantum particle features, and highlight the complexity of Shor's method with examples. Finally, this has attracted academics' interest in Quantum cryptography as a means of ensuring data security. Mavroeidis et al. [10] discusses the BB84 protocol on post quantum cryptography section, lattice-based cryptography, multivariate-based encryption, hash-based signatures, and code-based cryptography, among other quantum key distribution methods and mathematical-based solutions. Both traditional public key methods and symmetric key algorithms are at risk from quantum computers. Ekera & Håstad [11] summarized Ekera's quantum algorithm for computing short discrete logarithms in this paper to allow for various tradeoffs between the number of times the algorithm must be executed on one hand, and the algorithm's complexity and the requirements it imposes on the quantum computer on the other. We have also covered applications for methods that compute small discrete logarithms. When factoring an n-bit integer, Shor's approach uses an exponent of length 2n bits, but our algorithm uses slightly more than n/2 bits. This improvement is the result of a combination of two optimizations. Furthermore, by running the quantum process many times to generate a set of partial results, we gain a factor of two. In a traditional post-processing step, we retrieve the discrete logarithm d from this set.

IV. IMPLEMENTATION

The key insight of Shor's Algorithm is turning the problem of factorization into a period finding. This is based on the idea that large-number factorization is extremely hard to describe. It employs a method to determine the periodicity of things. Besides, this leads to a problem that threatens RSA Cryptosystem such they rely on "How hard is it to factorize a large number into two prime factors" this has exponential complexity in classical computers where the quantum computers take polynomial time. Period finding problem as shown in figure [1] says, "If we have periodic function how can we find period distance r?" This sounds interesting but how is Shor's algorithm linked to the factorization problem?

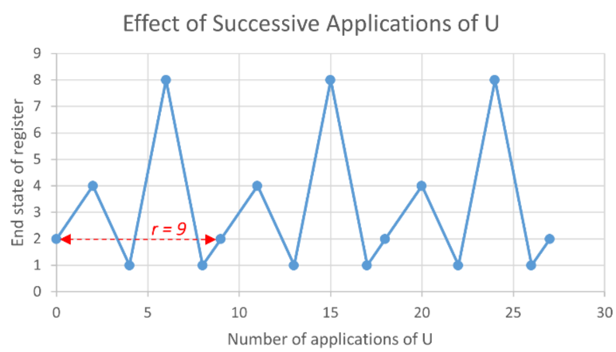


Figure 1| Periodic Finding Problem

We know from fundamental mathematics that

$$\boldsymbol{a^2 \equiv b^2 mod\ N}$$
$$\boldsymbol{a^2 - b^2 = N}$$
$$\boldsymbol{(a-b)(a+b) = N}$$
$$\boldsymbol{p \times q = N}$$
$$\boldsymbol{where\ GCD(a+b\,, N) = p\ \ and\ GCD(a-b, N) = q}$$

The interesting part is that if we have **b** as 1 it will reduce our problem to have a good guess of **a** only!

$$\boldsymbol{a^2 \equiv 1^2 mod\ N}$$

We need to choose an **a** such that it coprime with **N** and then find the order of **r** for this function

$$\boldsymbol{a^r\ mod\ N} \text{ (Finding period)}$$

and if **r** is even then we can apply

$$\boldsymbol{X \equiv a^{r/2} mod\ N}$$

after that **{p, q}** at least one of them contained in **gcd (x+1, N)** or **gcd (x-1, N)**, some of the **a** selection give us a partial solution, if **r** is odd you need to make another guess.

How can we use Shor's Algorithm to break RSA?

1) Pick a number "a" that is coprime with N
2) Find the order "r" of the function $a^r\ mod\ N$ (smallest r such that $a^r \equiv 1\ mod\ N$)
3) If r is even: **good news**
   $X \equiv a^{r/2} mod\ N$
   If $X + 1 \not\equiv 0\ mod\ N$: **good news**
   {p, q} at least one of them contained in gcd (x+1, N) or gcd (x-1, N)
   else: Find another "a"

The circuit will be implemented as shown in figure [2].

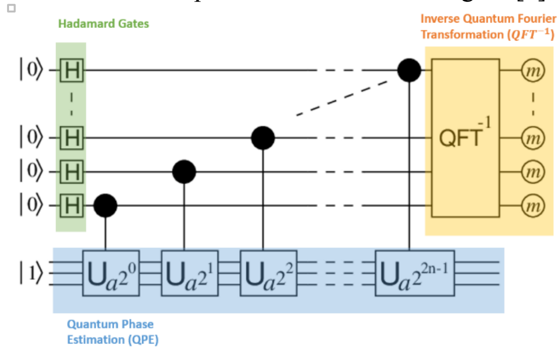


Figure 2 | Circuit Implementation

At the beginning, every qubit will connect to the **Hadamard gate** (single qubit gate) this will allow the qubit to be in a superposition state we can demonstrate mathematically with

$$\mathrm{H}(gate) = \frac{1}{\sqrt{2}} \times \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Example of applying Hadamard gate on vector $|\mathbf{0}\rangle$

$$\mathrm{H}|0\rangle = \frac{1}{\sqrt{2}} \times \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = |+\rangle$$

Also, in figure [3] we can see the output of H$|\mathbf{1}\rangle$ probability with 50% chance

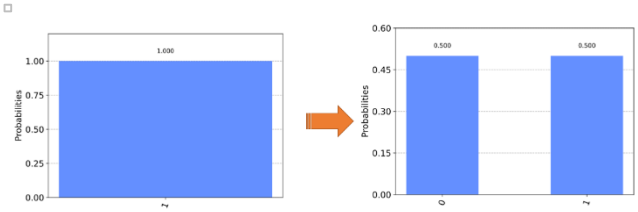


Figure 3 | Hadamard Gate in One Qubit

After applying the Hadamard gate we need to apply a Quantum Phase Estimation (QPE) to find the eigenvalue of a unitary operator. Briefly, phase estimation implies estimating the value of ϕ the eigenvalue. In Shor's Algorithm, our goal was to find the period **r** using some modulo function.

However, to execute it properly we must create a link between the period of a function and the phase value of the eigenvalue to find **r**, one approach to do this is using a Unitary operator after each Hadamard gate as in figure [4] below.

eigenvector

$$U|\psi\rangle = e^{2\pi i\phi}\,|\psi\rangle, 0 \leq \phi < 1$$

complex eigenvalue

Figure 4 | Unitary Operator

Before going to the last step, we must distinguish between Fourier and Computational basis. The **computational basis** will store the value of either 0 or 1 (up or down) as shown in figure [5], where the **Fourier basis** will be transformed between the Z-basis states $|0\rangle$ and $|1\rangle$ to the X-basis states $|+\rangle$ and $|-\rangle$ as shown figure [6].

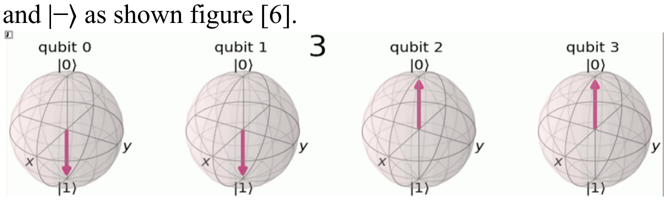


Figure 5 | Computational Basis

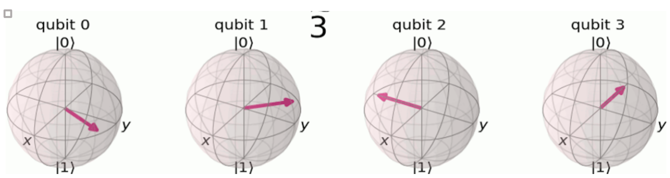


Figure 6 | Fourier Basis

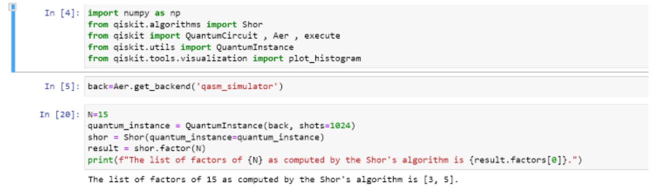
Now the output for all Hadamard gates will be stored on a Fourier basis therefore we must convert it to a Computational basis to be able to measure it and this is called Inverse Quantum Fourier Transformation

$$|Fourier\ State\rangle \xrightarrow[QFT^{-1}]{} |Computational\ \rangle$$

$$QFT^{-1}|\tilde{x}\rangle \rightarrow |\mathrm{X}\rangle$$

IBM provides a library that hides a lot of the complexity of implementing circuits in the IBM lab. We used python to load these libraries and run Shor's algorithm on Aer quantum circuit simulator [1].

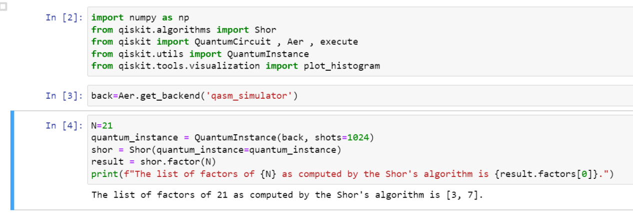
Case 1: Factor Number 15



```
In [4]: import numpy as np
        from qiskit.algorithms import Shor
        from qiskit import QuantumCircuit , Aer , execute
        from qiskit.utils import QuantumInstance
        from qiskit.tools.visualization import plot_histogram

In [5]: back=Aer.get_backend('qasm_simulator')

In [20]: N=15
         quantum_instance = QuantumInstance(back, shots=1024)
         shor = Shor(quantum_instance=quantum_instance)
         result = shor.factor(N)
         print(f"The list of factors of {N} as computed by the Shor's algorithm is {result.factors[0]}.")

         The list of factors of 15 as computed by the Shor's algorithm is [3, 5].
```

Figure 7 | Factoring 15

Case 2: Factor Number 21



```
In [2]: import numpy as np
        from qiskit.algorithms import Shor
        from qiskit import QuantumCircuit , Aer , execute
        from qiskit.utils import QuantumInstance
        from qiskit.tools.visualization import plot_histogram

In [3]: back=Aer.get_backend('qasm_simulator')

In [4]: N=21
        quantum_instance = QuantumInstance(back, shots=1024)
        shor = Shor(quantum_instance=quantum_instance)
        result = shor.factor(N)
        print(f"The list of factors of {N} as computed by the Shor's algorithm is {result.factors[0]}.")

        The list of factors of 21 as computed by the Shor's algorithm is [3, 7].
```

Figure 8 | Factoring 21

V. LIMITATION

Quantum computers are limited to qubit numbers, and they have issues with cooling and stability factors that run only under severe circumstances. Since qubits are the essential component in quantum computers, they are difficult to create and retain, specifically in large quantities. The biggest quantum computers contain several thousand qubits. In contrast, modern computers can have millions or billions of bits. Quantum computers are therefore incredibly tiny when compared to classical computers. Moreover, in terms of factoring numbers, modern classical computers can factor large prime numbers immediately, whereas Shor's algorithm is concerned with the Quantum Fourier Transformation operator, which can only determine factors of 21 qubits. Although a quantum computer will almost certainly never be a stand-alone system, it will be possible to break an RSA key in a reasonable amount of time when combined with a supercomputer.

VI. SUGGESTION

The National Institute for Standards and Technology (NIST) is working on post-quantum cryptography standardization. 59 schemas have been submitted to NIST and 7 of them reached the third round. We will look at the resulting standards in 2024[15]. But the interesting part is that 5 out of 7 of the third-round finalists rely on lattice-based cryptography for both public-key encryption and digital signatures. So, how about basing our security on lattice-based cryptography? What makes lattice cryptography hard for quantum computers to solve? Lattice-based cryptography relies on Short Vector. The problem is that for a given long basis lattice L, what is the shortest point to L? Here we have an amazingly simple example, lattice L, defined as all possible combinations of integer and basis. The basis is a set of stored vectors that can reproduce a new point. We have basis b1 and b2 given to the attacker. We asked them to solve the shortest vector to the origin of b1 and b2 (represented in the blue line). The answer will be the red line in figure 9[13].

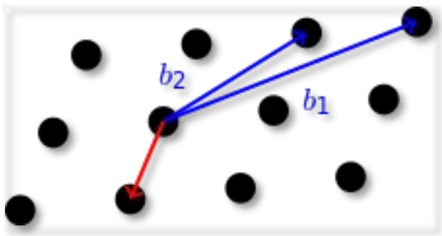


Figure 9 | lattice-based cryptography

To achieve success, some researchers devised an algorithm for determining an approximation factor and error correction. Moreover, some researchers have already adopted lattice-based signatures in their frameworks, such as the Internet-Of-Forensics Framework (IoF), in which they used lattice-based signcryption with programmable hash to ensure only legitimate devices are part of the framework, as it is assumed to be resistant to quantum attack [12].

VII. Conclusion

The paper discusses quantum computing methods in considerable detail and how Shor's algorithm can break the RSA algorithm. It has shown how Shor's algorithm puts secrecy and authentication services at risk. The key insight of Shor's algorithms is that they turn the problem of factorization into period findings and reduce the time complexity from exponential into polynomial time. The factoring issue can be solved on a traditional computer, and the results are validated using an IBM simulator. There are certain limitations since registers in typical computers cannot be in superposition. The experimental examination of quantum Shor's algorithms and RSA cryptosystems through the usage of IBM quantum experiences is undertaken as a proof of concept for factorizing integers of short length. One concern with Shor's algorithm is that the Quantum Fourier Transformation operator is compatible with the factoring technique, which can currently determine factors of equations up to 21 qubits. As a result, it cannot be factored into larger numbers [16].

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